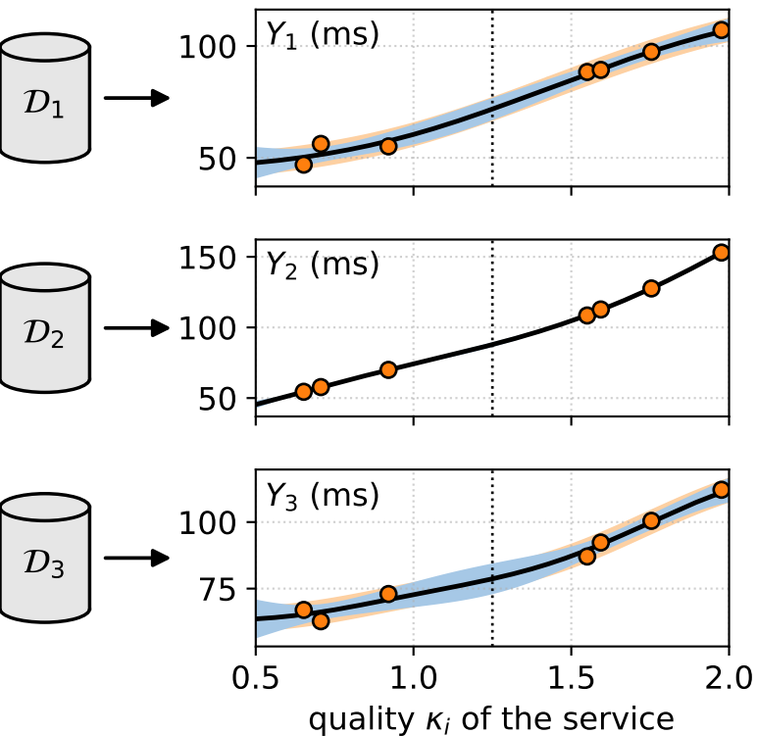
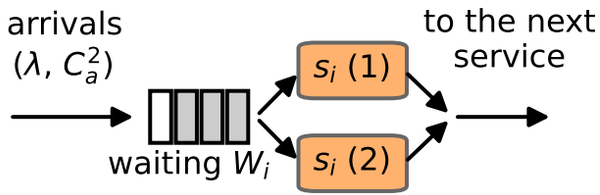
Inputs: composition $S$ configurations $\mathbf{x} = (x_1, \ldots, x_K)$, $x_i = \langle r_i, \kappa_i \rangle$ workers $c_i$ arrival process $(\lambda, C_a^2)$ violation probability $\varepsilon$ confidence $\gamma$

## ① Learning: one GP per service



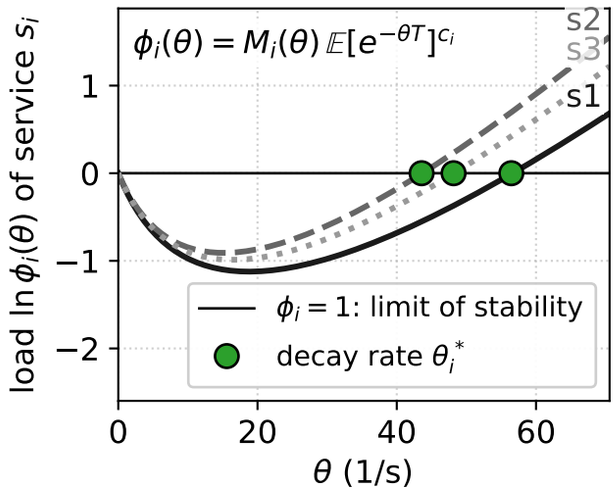

## ② Execution time and load



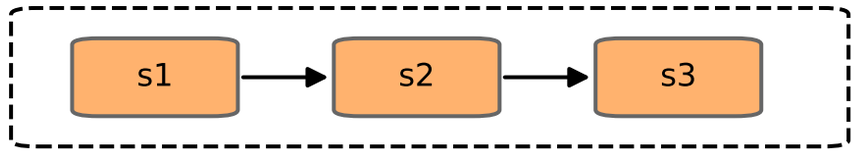

$c_i = 2$ workers

execution $Y_i \sim \mathcal{N}(m_i^+, \sigma_{al,i}^2)$

pessimistic mean $m_i^+ = m_i + z_\gamma \sigma_{ep,i}$




## ③ Composition into the PPG




*target pipeline: never executed*

Node: $d_1(\varepsilon/K)$ of s1

